EEE104 – Digital Electronics (I)

Lecture 13

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Emerging Electronics and Energy Technologies Research Centre (EEET)

Office: SC242

Office hour: 1:00-5:00 pm, Wednesday

In This Session

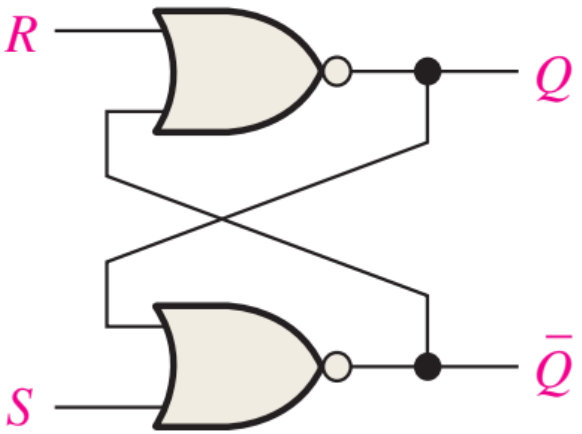
Flip-Flops and Related Devices

- **Latches**
- Edge-Triggered Flip-Flops
- Flip-Flop Applications

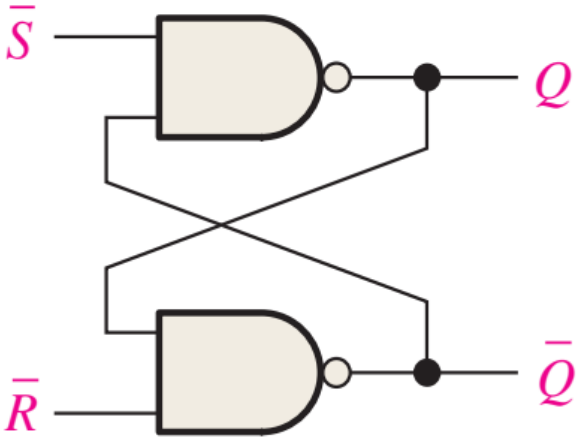
This session: basic S-R, gated S-R and gated D latches.

Latches – The Set-Reset (S-R) Latch

- A latch is a bistable digital circuit used for **storing** a bit.
- The output of one gate is fed back to the input of the other.



(a) Active-HIGH input S-R latch

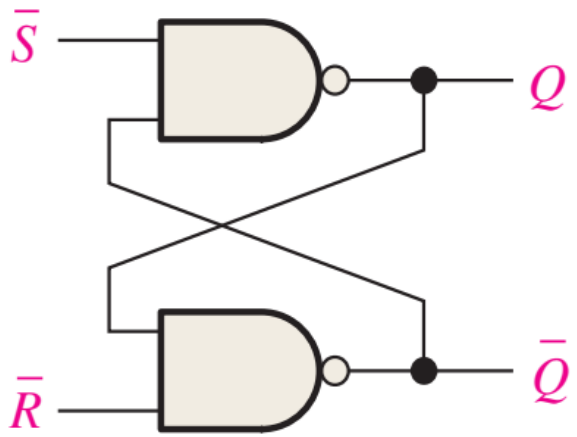


(b) Active-LOW input $\bar{S}$-$\bar{R}$ latch

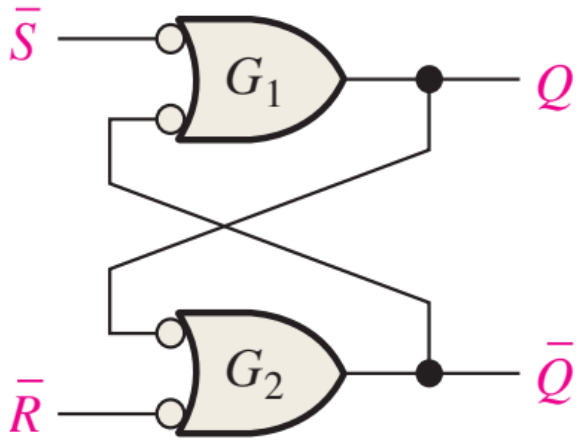
Bistable: **two stable states**, Q = 0 or Q = 1.

Latches – Active-Low S-R Latch

- A NAND gate is equivalent to a negative-OR gate.
- Hence the S-R latch equivalent implementation.
- It will be used extensively later.

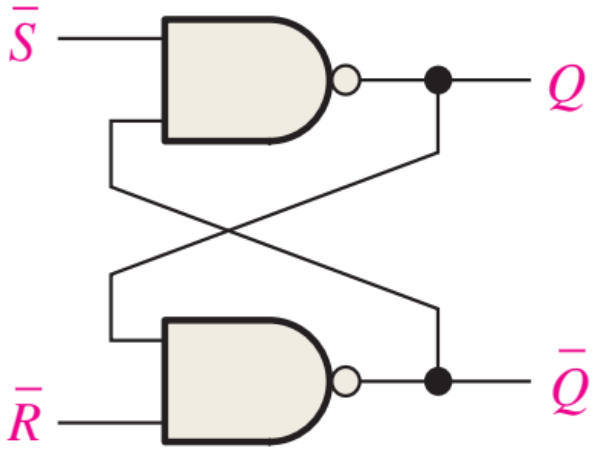


Active-LOW input S-R latch



Negative-OR equivalent

Latches – Active-Low S-R Latch



Upper gate after release:

$Q = \overline{1 \cdot 0} = 1$

Set operation

How does it work?

1. Suppose that $\overline{S}$ and $\overline{R}$ are both 1 originally.

2. When a negative pulse appears at $\overline{S}$, i.e. $\overline{S} = 0$, then $Q = 1$, $\overline{Q} = 0$

3. When the pulse finishes, i.e. $\overline{S} = 1$, Q is still 1 and $\overline{Q}$ is still 0.

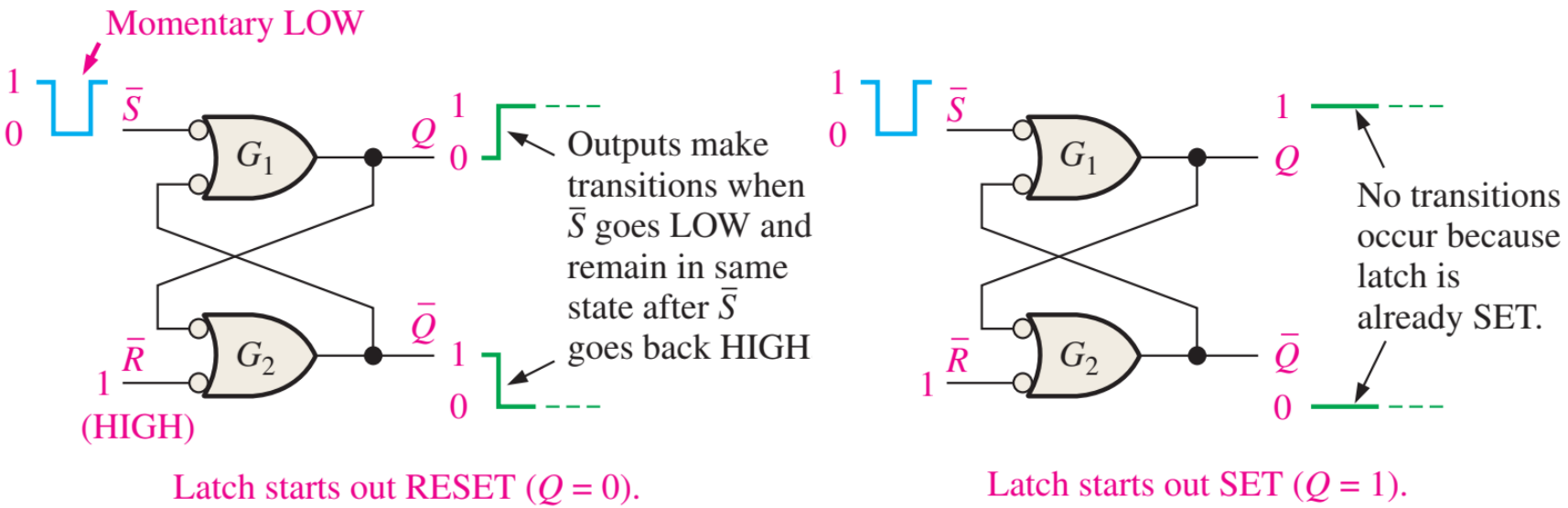
So the low-level pulse at $\overline{S}$ sets the output Q to 1.

Latches – Active-Low S-R Latch

1. When a negative pulse appears at $\overline{S}$, i.e. $\overline{S}$ = 0, then Q = 1, $\overline{Q}$ = 0

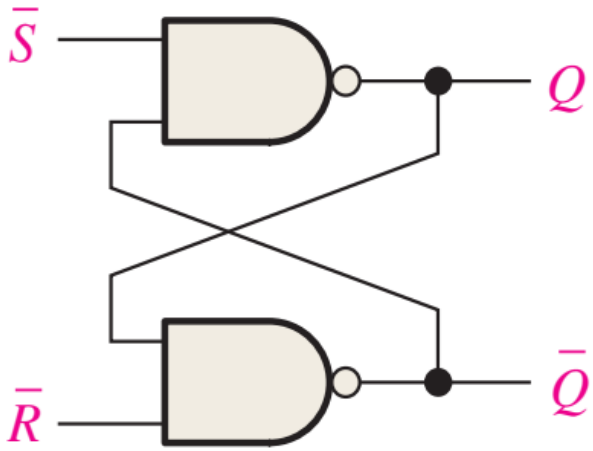
2. When the pulse finishes, i.e. $\overline{S}$ = 1, Q is still 1 and $\overline{Q}$ is still 0.

Two initial states for the SET operation



SET means Q = 1, for either initial state.

Latches – Active-Low S-R Latch



Lower gate after release:

$\overline{Q} = \overline{1 \cdot 0} = 1$

Reset operation

How does it work?

1. Suppose that $\overline{S}$ and $\overline{R}$ are both 1 originally.

2. When a negative pulse appears at $\overline{R}$, i.e. $\overline{R} = 0$, then $\overline{Q} = 1$, and $Q = 0$.

3. When the pulse finishes, i.e. $\overline{R} = 1$, Q is still 0 and $\overline{Q}$ is still 1.

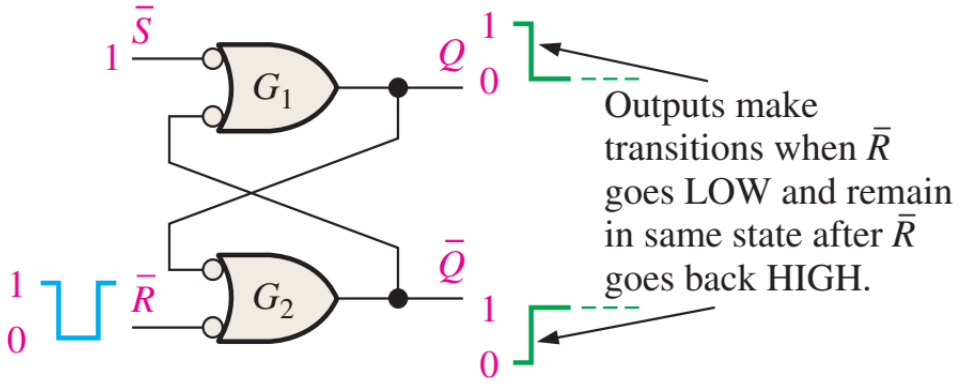
So the low-level pulse at $\overline{R}$ resets Q to 0.

Latches – Active-Low S-R Latch

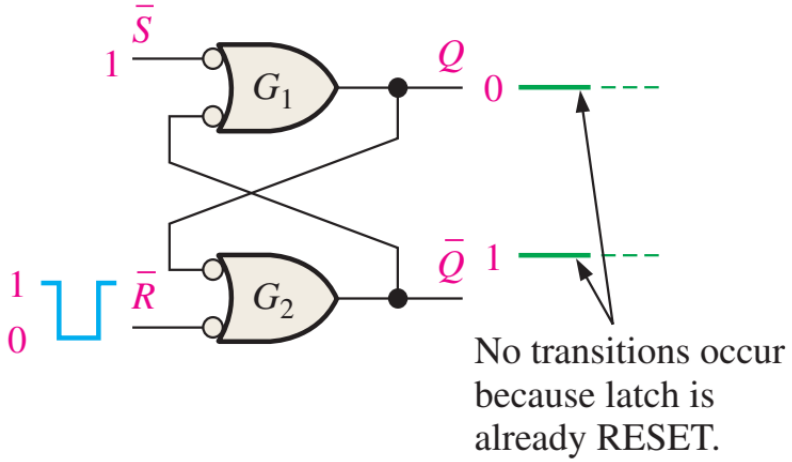
1. When a negative pulse appears at $\overline{R}$, i.e. $\overline{R}$ = 0, then $\overline{Q}$ 1 = and Q = 0.

2. When the pulse finishes, i.e. $\overline{R}$ = 1, Q is still 0, and $\overline{Q}$ is still 1.

Two initial states for the RESET operation



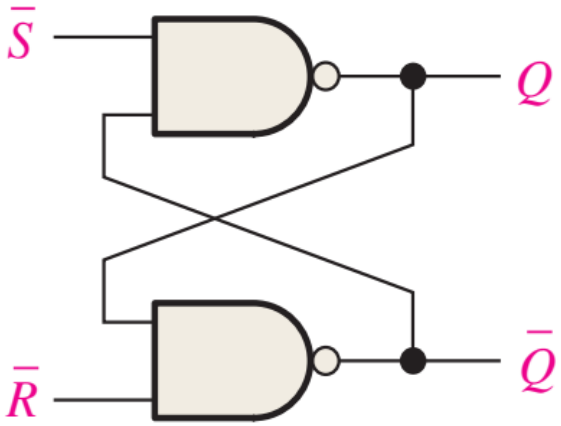
Latch starts out SET ($Q = 1$).



Latch starts out RESET ($Q = 0$).

RESET means Q = 0, for either initial state.

Latches – Active-Low S-R Latch



No-change condition

How does it work?

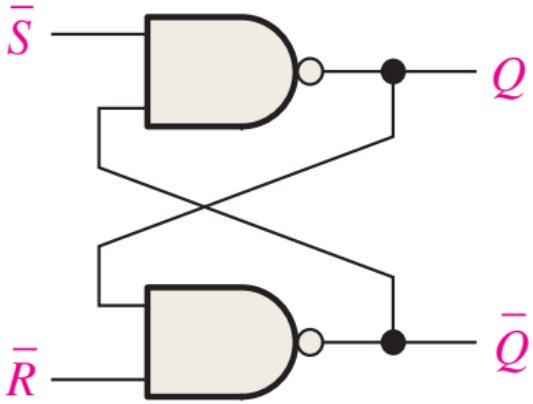
When $\overline{S} = \overline{R} = 1$, the outputs will not be changed.

$$Q = \overline{1 \cdot \overline{Q}} = \overline{\overline{Q}} = Q$$

$$\overline{Q} = \overline{1 \cdot Q} = \overline{Q}$$

HOLD preserves the previous state: 0 stays 0, 1 stays 1.

Latches – Active-Low S-R Latch

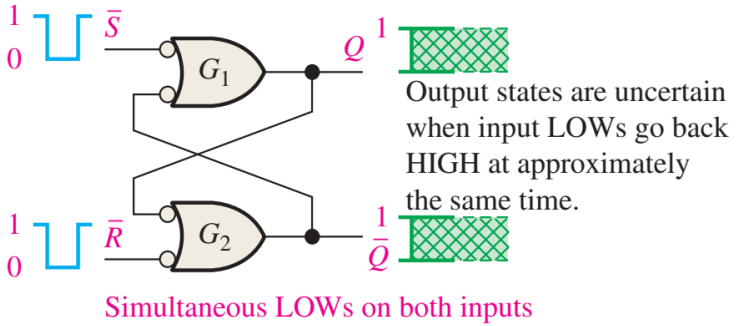


Invalid condition

How does it work?

What will happen when both $\overline{S}$ and $\overline{R}$ are LOW?

- **Both Q and $\overline{Q}$ will be forced HIGH.**
- When $\overline{S}$ and $\overline{R}$ become HIGH simultaneously, both outputs will attempt to go LOW. The next state cannot be reliably predicted because of gate-delay differences.

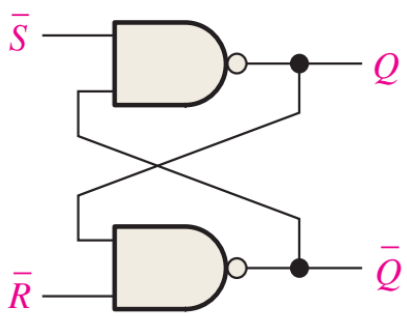


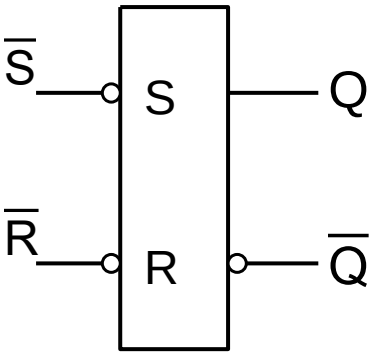
Simultaneous LOWs on both inputs

Both LOW: Q = $\overline{Q}$ = 1.
Released together: state uncertain.

Latches – Active-Low S-R Latch

An S-R latch with **active-LOW** inputs is formed using **NAND** gates.





$\overline{S}$	$\overline{R}$	Q	$\overline{Q}$	Comments
1	1	NC	NC	No change. Present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	**0**	**1**	**1**	**Invalid condition**

NC: No change. Latch remains in present state.

Practice 1: Active-Low S-R Latch

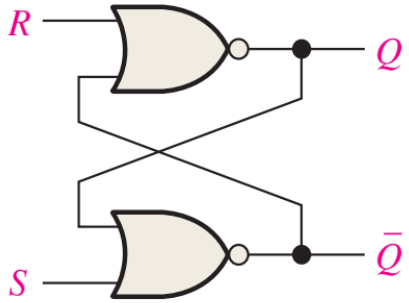
NAND S-R latch with active-LOW inputs. Initial Q = 0.

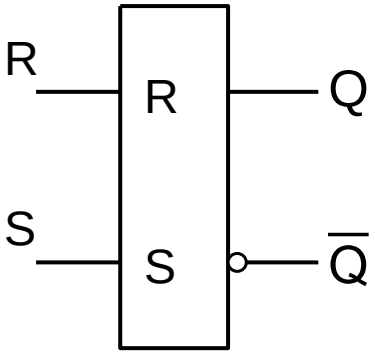
Apply each pair ($\overline{S}$, $\overline{R}$) in order. Write Q after each step.

Step	**1**	**2**	**3**	**4**	**5**
$\overline{S}$	1	0	1	1	1
$\overline{R}$	1	1	1	0	1
Q after	?	?	?	?	?

Latches – Active-High S-R Latch

An S-R latch with **active-HIGH** inputs is formed using **NOR** gates.





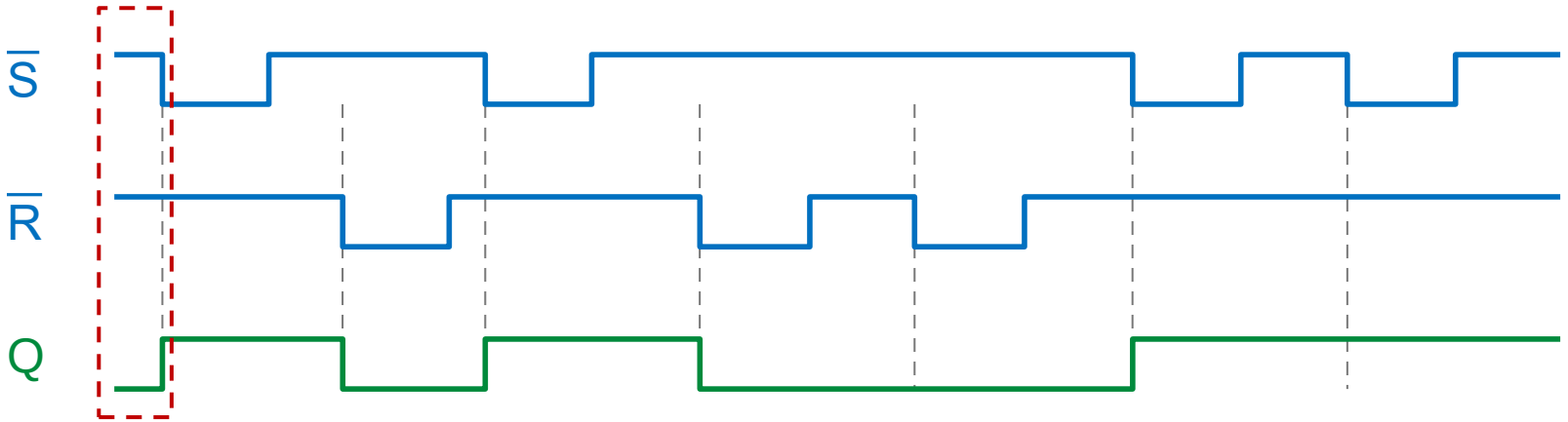
S	R	Q	$\overline{Q}$	Comments
0	0	NC	NC	No change. Present state.
0	1	0	1	Latch RESET.
1	0	1	0	Latch SET.
1	**1**	**0**	**0**	**Invalid condition**

Check input polarity before using a truth table.

Latches – The S-R Latch

An Example: For a latch with active-LOW inputs

Initial Q = 0



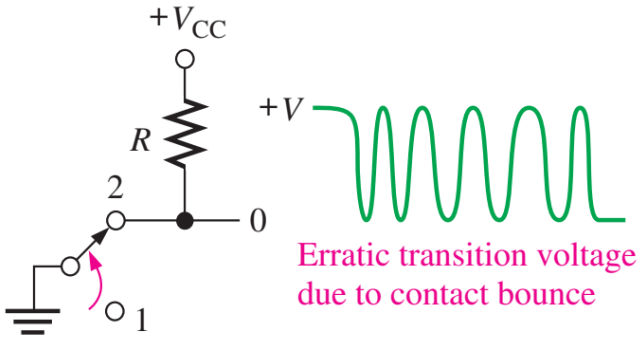
Initial states

$\overline{S}$	$\overline{R}$	Q	$\overline{Q}$	Comments
1	1	NC	NC	No change. Present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	**0**	**1**	**1**	**Invalid condition**

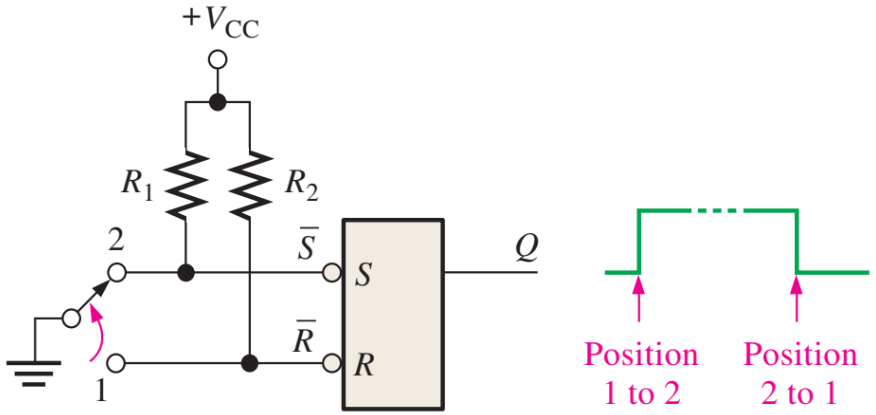
Latches – The S-R Latch

Application Example

The latch as a contact-bounce eliminator



(a) Switch contact bounce

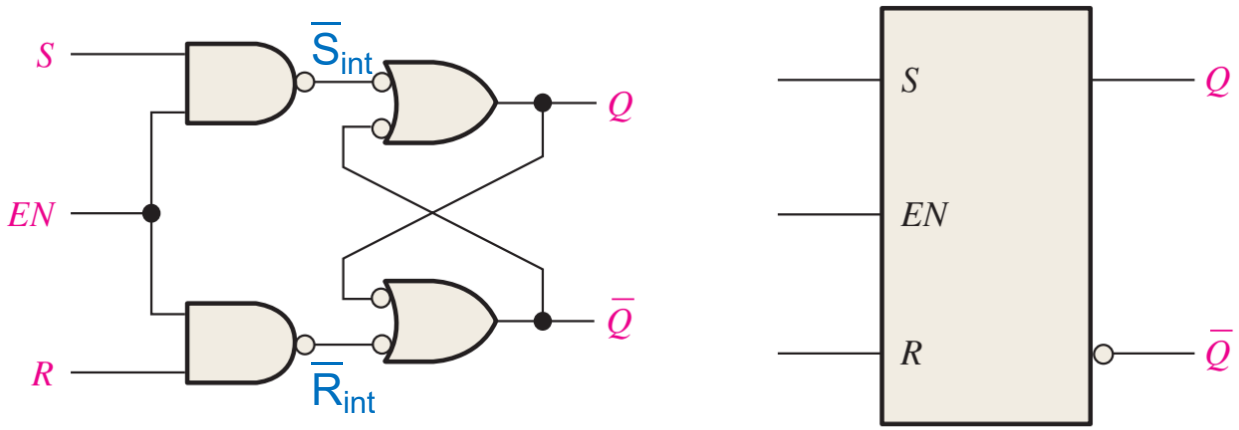


(b) Contact-bounce eliminator circuit

Switch	$\overline{S}$	$\overline{R}$	Q
1 to 2	0	1	1 (SET)
2 to 1	1	0	0 (RESET)
Between contacts	1	1	NC

Latches – The Gated S-R Latch

- A gated latch is a latch with an enable input EN.
- When EN is LOW, the latch will not change.
- When EN is HIGH, the output is controlled by the state of the S and R inputs.



(a) Logic diagram (b) Logic symbol

The gated latch is a level-sensitive device.

Gated S-R Latch: Enable and Feedback

Inputs to the NAND core:

$\overline{S}_{int} = \overline{S \cdot EN}$ $\qquad$ $\overline{R}_{int} = \overline{R \cdot EN}$

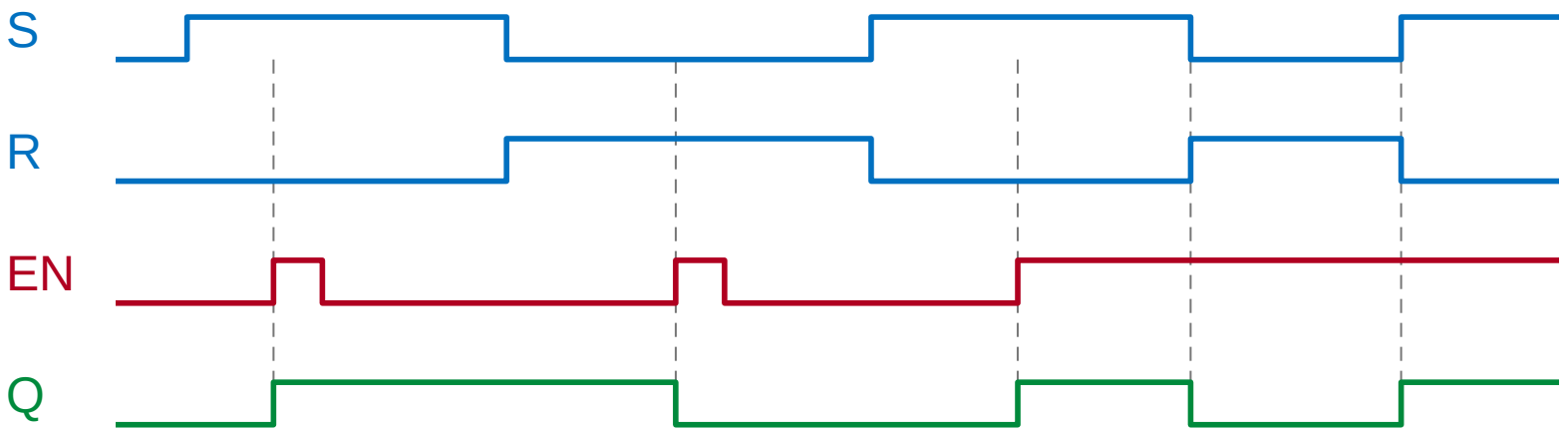
EN	S	R	$\overline{S}_{int}$	$\overline{R}_{int}$	Q	$\overline{Q}$	Action
0	0	0	1	1	NC	NC	Hold
0	0	1	1	1	NC	NC	Hold
0	1	0	1	1	NC	NC	Hold
0	1	1	1	1	NC	NC	Hold
1	0	0	1	1	NC	NC	Hold
1	1	0	0	1	1	0	Set
1	0	1	1	0	0	1	Reset
1	**1**	**1**	**0**	**0**	**1**	**1**	**Invalid**

EN = 0: HOLD. EN = 1: external S and R are active-HIGH.

Latches – The Gated S-R Latch

An Example: For a latch with active-HIGH inputs

Initial Q = 0

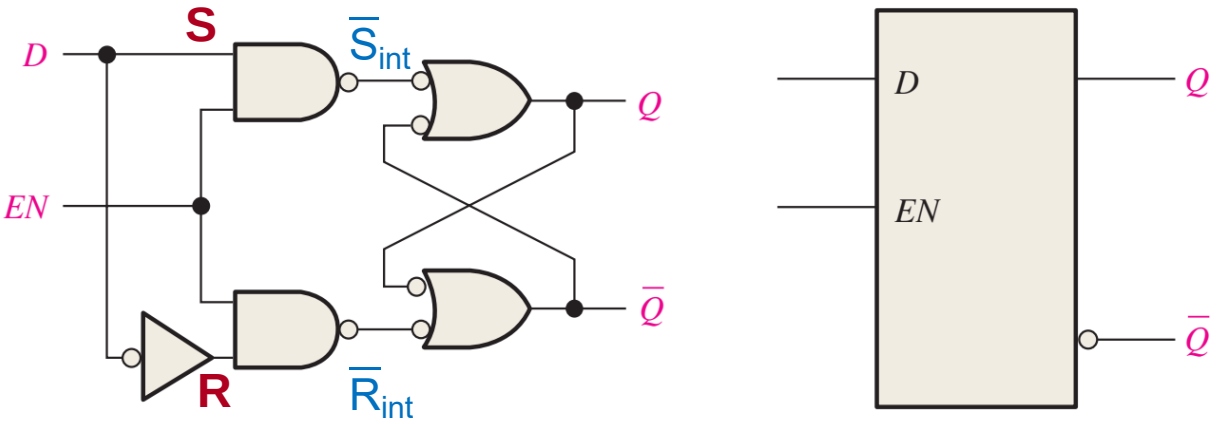


While EN = 1:

S	R	Q	$\overline{Q}$	Comments
0	0	NC	NC	No change
0	1	0	1	Latch RESET.
1	0	1	0	Latch SET.
1	**1**	**1**	**1**	**Invalid condition**

Latches – The Gated D Latch

- It has only one input in addition to EN.
- When D = 1 and EN = 1, Q = 1.
- When D = 0 and EN = 1, Q = 0.
- Output Q = D when EN is HIGH.



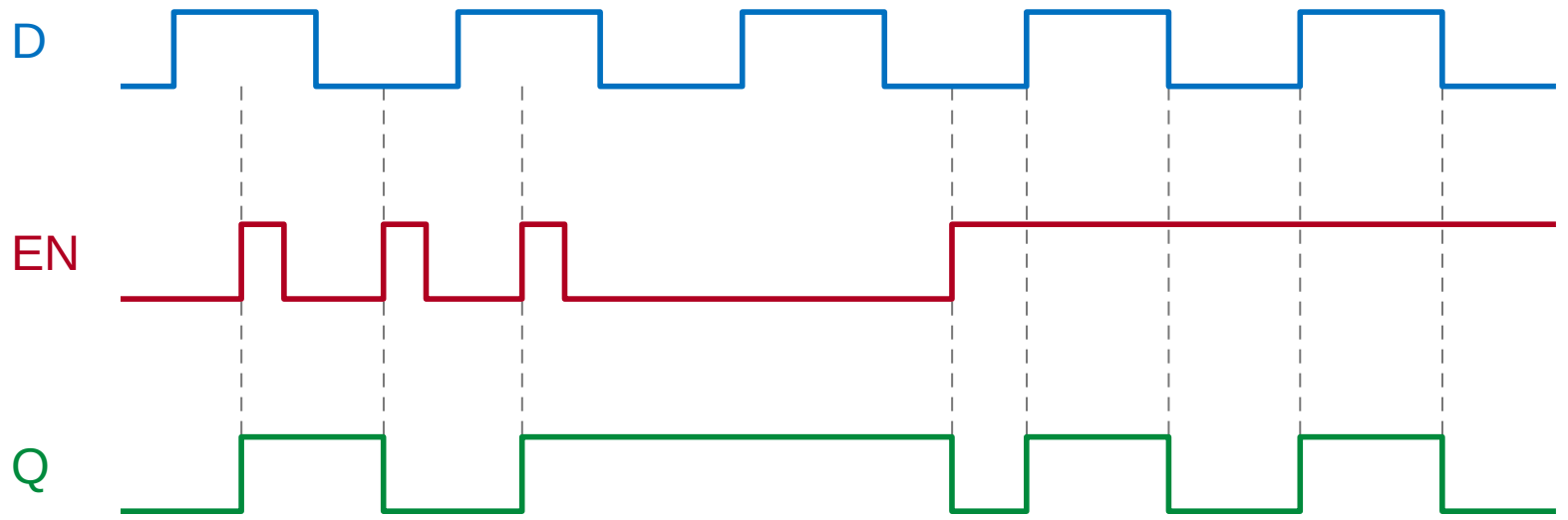
(a) Logic diagram (b) Logic symbol

S = D, R = $\overline{D}$: the two data controls are complementary.

When EN is LOW, D does not affect the state of the latch.

Latches – The Gated D Latch

An Example: Initial Q = 0



EN HIGH: Q follows D. **EN LOW**: Q holds its state.

Practice 2: Gated D Latch

Gated D latch. Initial Q = 0.

Apply each (EN, D) pair in order. Write Q after each step.

Step	**1**	**2**	**3**	**4**	**5**
EN	0	1	1	0	1
D	1	1	0	1	1
Q after	?	?	?	?	?